LAB 02B - Marked Example

October 26, 2026

```
[ ]: # In Python it is standard practice to import the modules we need at the very␣
      ↪top of our scripts
     import numpy as np
     import matplotlib.pyplot as plt
```

1 Exercise : Bringing Everything Together (5 Marks)

In exercise 2 we used variables to hold single values for speed, time and distance in our projectile motion scenario. In that sense we used Python like a glorified calculator; however, the beauty of programming languages is that they can remove the effort needed for repetitive calculations and instead of calculating the velocity at just one or two points we could calculate the velocity at hundreds of points and track it across the projectile's flight.

In this exercise you will revisit the same problem, but plot `velocity_x`, `velocity_y` and `velocity_mag` for the entire flight. Likewise, you will plot the body altitude and distance from launch. You should refer back to exercise 2 and your earlier code… which hopefully was clearly commented!

As a reminder, we're working with the following scenario: A body is launched into the air at an angle $\theta = 25^\circ$ (compared to ground level) with initial velocity of $v_0 = 7.5\text{m/s}$. Assume an acceleration due to gravity of $g = 9.81\text{ms}^{-2}$.

You should assign the input numbers to suitable variables, and define a suitable array containing a range of values representing the time-axis, starting at `t=0` and covering the total time that the body is above ground (see exercise 2). Your code should create new arrays containing the horizontal and vertical velocity, speed and horizontal and vertical position.

Plot the following as four subplots within a single figure (refer back to the examples for `subplot` if you need help):

1. the instantaneous horizontal velocity `velocity_x` and vertical velocity `velocity_y` (on one set of axes) for the entire flight,
2. the magnitude velocity `velocity_mag` (on a separate set of axes) for the entire flight,
3. the instantaneous altitude `position_y` and horizontal displacement from launch `position_x` (on one set of axes),
4. the actual projectile, e.g. the altitutde as a function of horizontal displacement (on a separate set of axes) from launch point of the body for the entire flight.

Save your figure as a file named `projectile_motion.png`.

You will need to calculate the total time the projectile is airborne in order to know the appropriate range of times to consider. Reality check: think about whether the graphs you have plotted make sense. Why is there a minimum in the speed?

Ensure you: comment your code, use informative names for your variables, print your answers to an appropriate number of significant figures, think about whether your results look right.

Tip: your plot will probably come out rather small because four plots are squeezed into the space normally occupied by one. You may find the following code helpful:

```
plt.figure(figsize=[10,10])  #  in units of Inches!
```

[ ]:
```
# Write your set-up and calculations for question 1
```

[ ]:
```
# Write your calculations for question 2 here
```

[ ]:
```
# Make your plots here
```